

Generative AI for Healthcare

Causality in Medical Imaging

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Generative AI: Recent Progress

Prompt: A crab made of cheese on a plate



Stable Diffusion (Stability AI)



DALL·E (OpenAI)

Generative AI: Recent Progress

Prompt: An old rusted robot wearing pants and a jacket riding skis in a supermarket



Stable Diffusion (Stability AI)



DALL·E (OpenAI)

Generative AI: Recent Progress

Prompt: A whimsical and creative image depicting a hybrid creature that is a mix of a waffle and a hippopotamus...



Stable Diffusion (Stability AI)



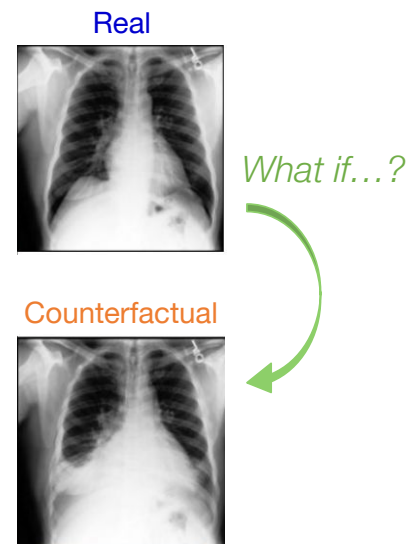
DALL·E (OpenAI)

Generative AI: For Medical Imaging!



“What would this **medical scan** have looked like if the patient had a **disease**?”

- **Understanding the data-generating process:** Move beyond correlation by answering “*what if?*” questions
- **Bias auditing:** Stress-testing AI models on different patient demographics (e.g. *age*, *race* etc)
- **AI interpretability:** Help explain which features drive decisions in AI systems
- **Synthetic data:** Improve AI models by augmenting datasets with synthetic examples



What is Causality?

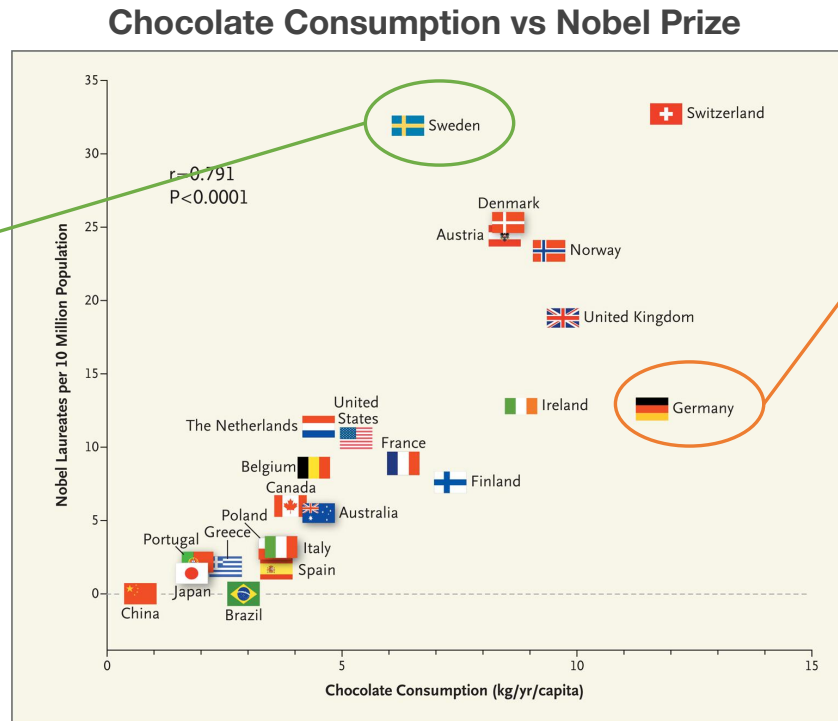
- Scientific inquiry is often motivated by **causal** questions:
 - How effective is a **treatment** in preventing a **disease**?
 - If I take this **pill**, will my **headache** be gone?
 - Would my **grades** have been better had I **studied** more?

Causality is the relationship between **cause** and **effect**

Example: chocolate

"Correlation does *not* imply causation"

Secretly the best chocolate?



Least effective chocolate?

Messerli. Chocolate Consumption, Cognitive Function, and Nobel Laureates. N Engl J Med 2012

Inspired by Jonas Peters

Reichenbach's Common Cause Principle



Hans Reichenbach

1. Chocolate consumption causes Nobel prize win

$$C \rightarrow N$$

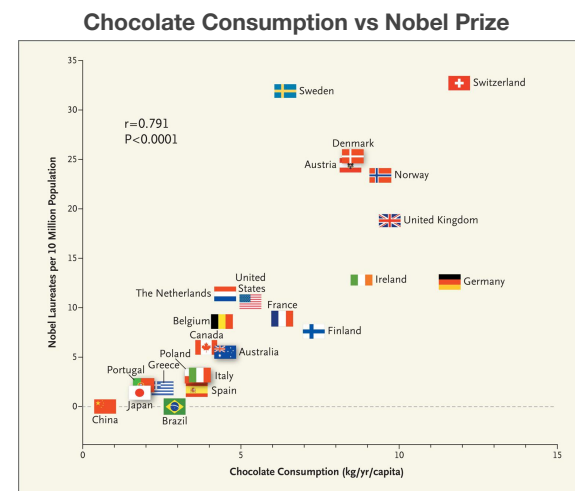
2. Nobel prize win causes chocolate consumption

$$C \leftarrow N$$

3. Both are caused by an unknown factor U

$$C \leftarrow U \rightarrow N$$

Could U be GDP per capita or wealth per adult?



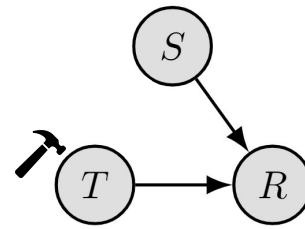
Messerli. Chocolate Consumption, Cognitive Function, and Nobel Laureates. N Engl J Med 2012

Simpson's Paradox

→ Stone size (S) is a **confounder**

→ Make treatment (T) independent of size:

- ◆ What's the recovery (R) rate if **all** subjects receive treatment **A** vs **B**?



Edward Simpson

Kidney Stone Size	Treatment A	Treatment B
Small	93% (81/87)	87% (234/270)
Large	73% (192/263)	69% (55/80)
Both	78% (273/350)	83% (289/350)

??

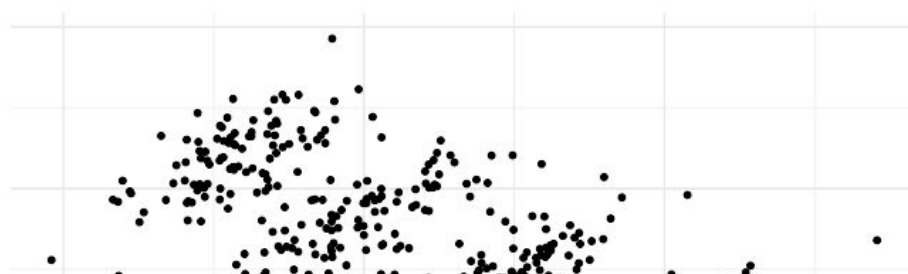
Patients with **large stones** received the **better treatment (A)**, and those with **small stones** received the **inferior treatment (B)**

Charig CR, *et al.* Comparison of treatment of renal calculi by open surgery, percutaneous nephrolithotomy, and extracorporeal shockwave lithotripsy. Br Med J (Clin Res Ed). 1986 Mar 29.

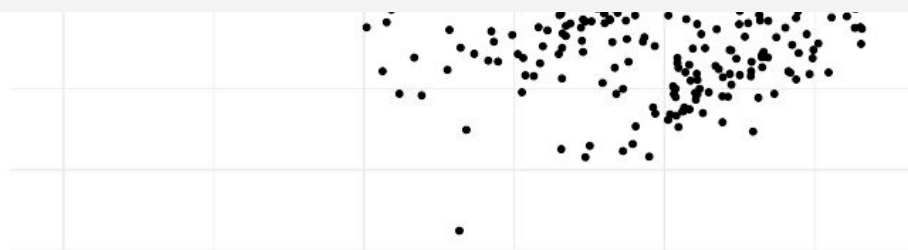
Simpson's Paradox



Edward Simpson



Correlations may **reverse** depending on how we aggregate data and its subpopulations



Predictive Modelling

Given an image X , train a model to predict some label Y

$$P(Y|X)$$

→ Assumptions:

- ◆ Sufficient training data (X, Y) is available
- ◆ Training and test data come from the same distribution

Predictive Modelling: A Causal Perspective

What is the causal relationship between image X and label Y ?

$$P(Y|X)$$

$$X \rightarrow Y$$

causal

(predict **effect** from **cause**)

$$X \leftarrow Y$$

anti-causal

(predict **cause** from **effect**)

Example: Skin Lesion Classification

X – dermoscopic image

Y – biopsy-derived diagnosis

$$X \leftarrow Y$$

anti-causal
(predict **cause** from **effect**)

$$P(Y|X)$$

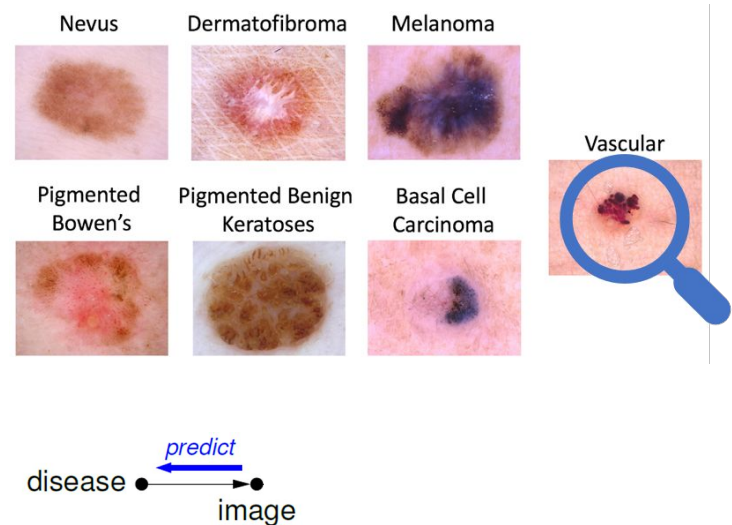


Image: ISIC 2018 challenge (<https://challenge2018.isic-archive.com/task3/>)

Example: Brain Tumour Segmentation

X – structural brain MRI

Y – manually drawn contour

$$X \rightarrow Y$$

causal
(predict **effect** from **cause**)

$$P(Y|X)$$

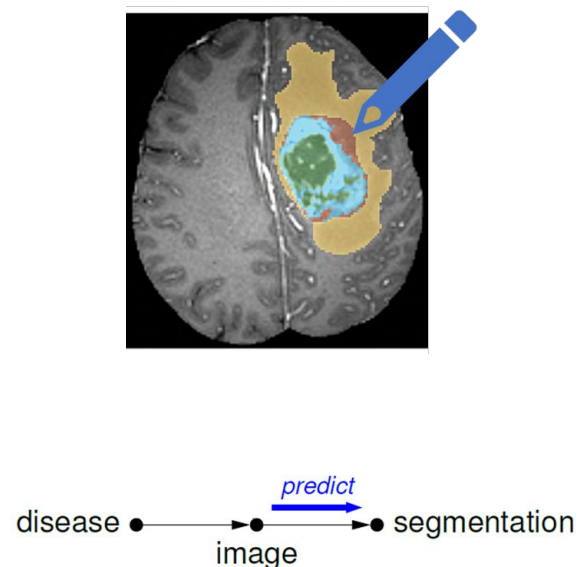


Image: ISIC 2018 challenge (<https://challenge2018.isic-archive.com/task3/>)

Example: Radiology Reports

X – chest X-ray

Y – diagnosis extracted from report

$$X \overset{?}{\leftrightarrow} Y$$

causal or anti-causal?

Not clear whether the diagnosis was derived from the image or another lab result...

$$P(Y|X)$$

Meta-information is needed to establish causal relationships

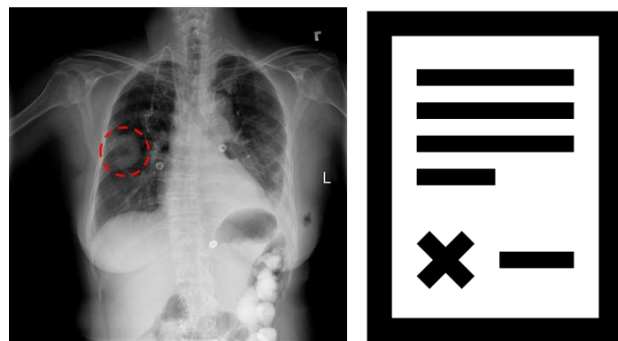
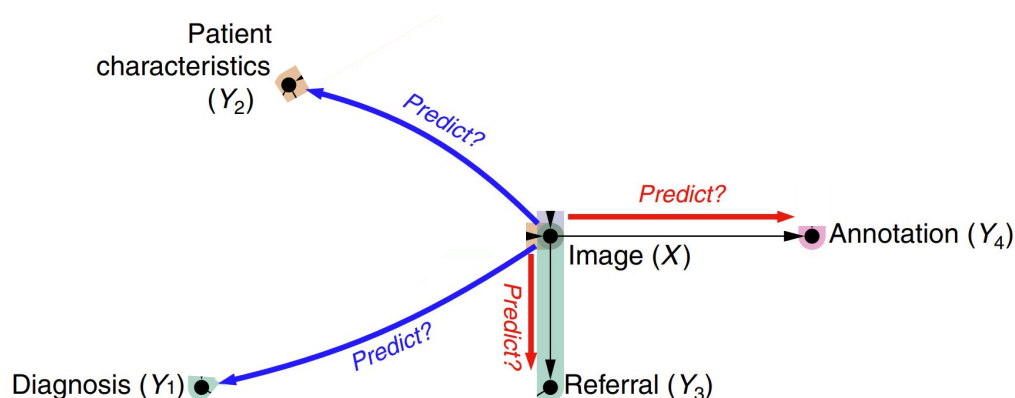
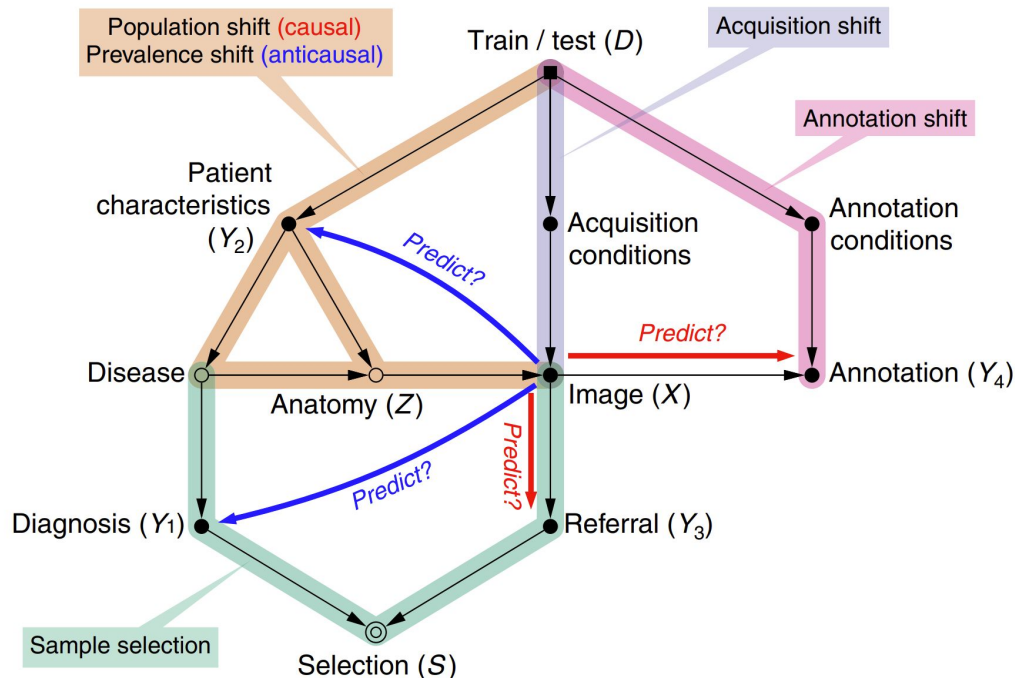


Image: ISIC 2018 challenge (<https://challenge2018.isic-archive.com/task3/>)

The Data Generating Process



The Data Generating Process



Castro, D. C., Walker, I., & Glocker, B. (2020). Causality matters in medical imaging. *Nature Communications*, 11(1), 3673.

Ladder of Causation

Counterfactuals

What if X had happened?

- I. Counterfactual reasoning
- II. Deduce causes for observed events

Intervention

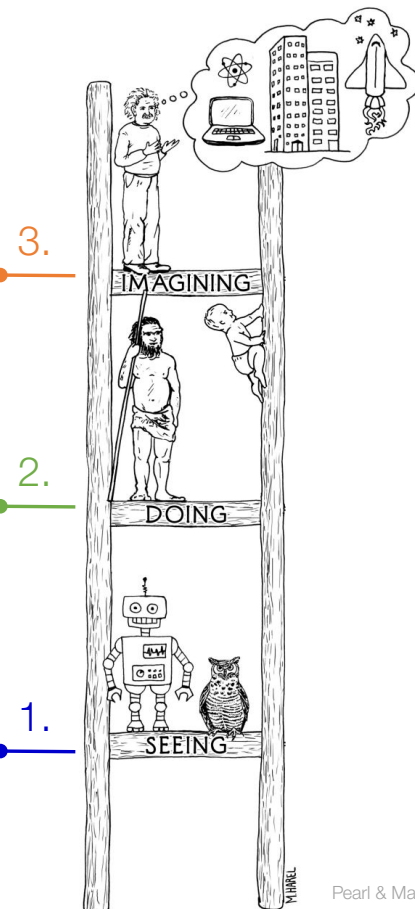
What will Y be if i **do** X ?

- I. Use experiments to identify causal effects
- II. Crucial for planning and policy making

Association

How would **seeing** X change my belief in Y ?

- I. Modelling correlations $P(Y|X)$
- II. Modern AI excels at this task

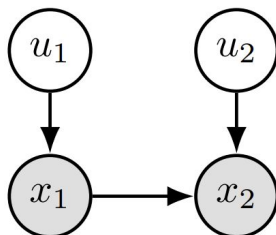


Judea Pearl

Structural Causal Models

→ **Example:** A simple SCM:

- ◆ x_1, x_2 are **endogenous** whereas u_1, u_2 are **exogenous**

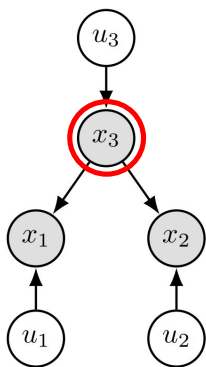


$$x_1 := f_1(u_1), \quad u_1 \sim \mathcal{N}(0, 1)$$

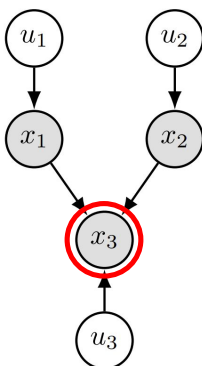
$$x_2 := f_2(x_1, u_2), \quad u_2 \sim \mathcal{N}(0, 1)$$

Structural Causal Models

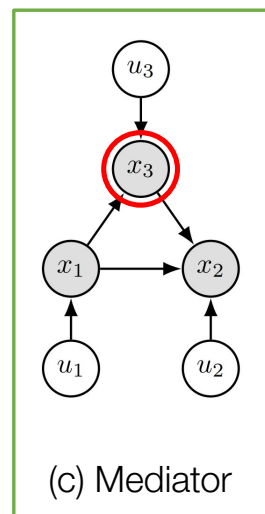
→ Acyclic SCMs can be represented by a **Directed Acyclic Graphs (DAGs)**, with edges pointing from **causes** to **effects**



(a) Confounder



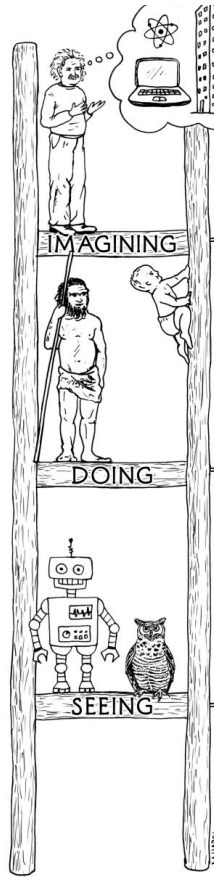
(b) Collider



(c) Mediator

Counterfactuals

(rung 3.)



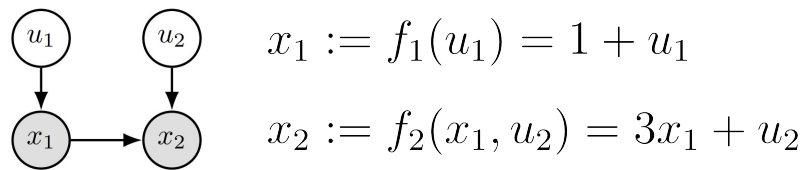
→ Causal models can consider hypothetical scenarios:

- *“Arrived late, what if the train had been on time?”*

→ Counterfactual inference involves three steps:

1. **Abduction:** Infer why I was late (e.g. delayed train)
2. **Action:** Imagine the train was on time
3. **Prediction:** Predict what time I would have arrived

Example: Computing Counterfactuals



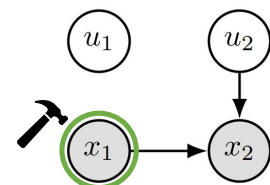
Q: Given we observed $\{x_1=2, x_2=4\}$, what would x_2 have been had x_1 been 5?

1. Abduction: $x_1 := 2 = 1 + u_1 \implies u_1 = 1$

$$x_2 := 4 = 3 \cdot 2 + u_2 \implies u_2 = -2$$

2. Action: $\tilde{x}_1 := 5$

3. Prediction: $\tilde{x}_2 = 3\tilde{x}_1 + u_2 = 3 \cdot 5 - 2 = 13$



Generative AI for Causal Modelling



- Use generative AI to learn causal models
- Estimate **counterfactuals**, i.e. answer “*what if?*” questions
- **Challenging** in complex problems, e.g. medical imaging

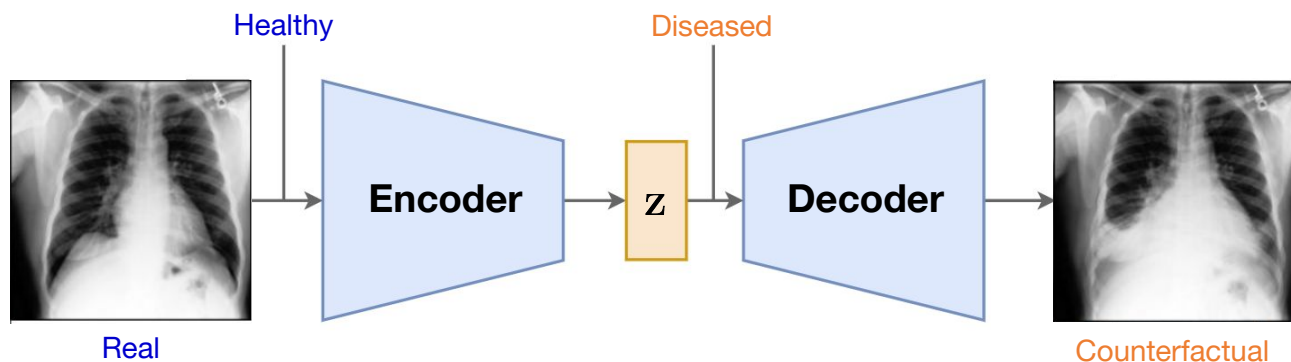
Research Questions:

- Can we generate plausible counterfactuals of medical images?
- How do we evaluate them?

Pawlowski, Castro, Glocker. Deep Structural Causal Models for Tractable Counterfactual Inference. *NeurIPS* 2020

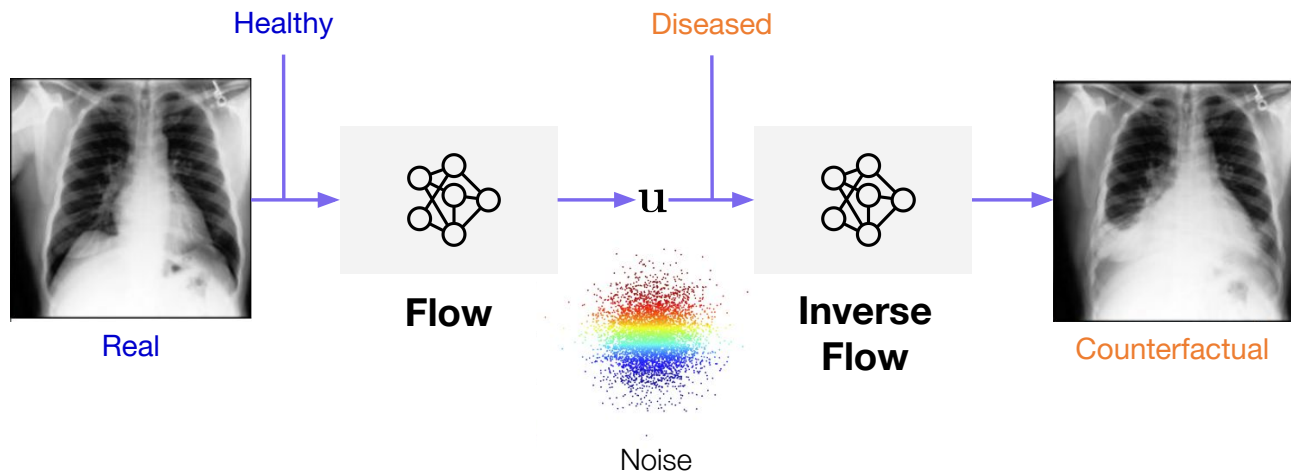
Causal Mechanism: Variational Autoencoder

- **High-level Example:** computing an image counterfactual



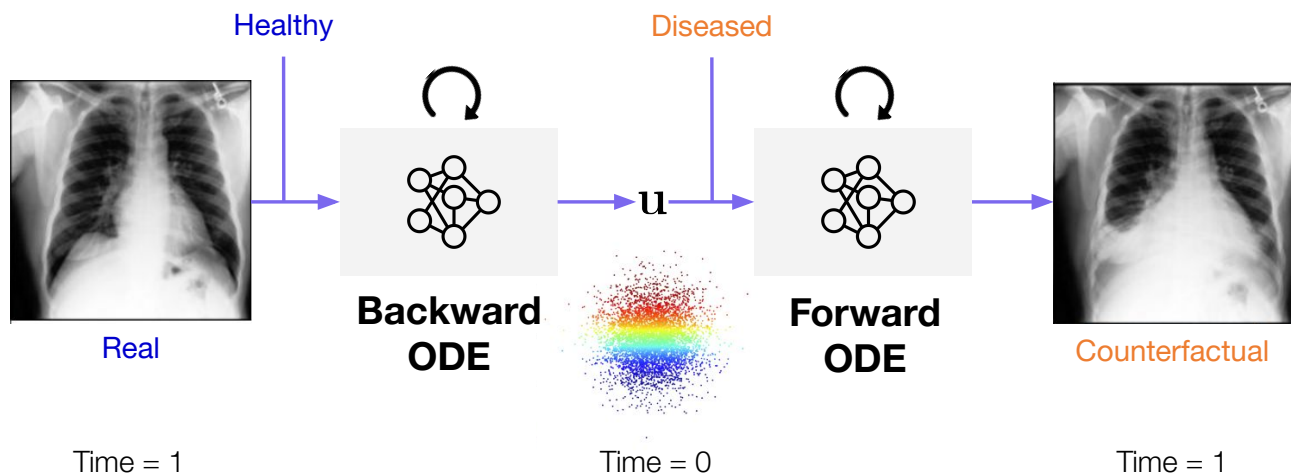
Causal Mechanism: Normalizing Flow

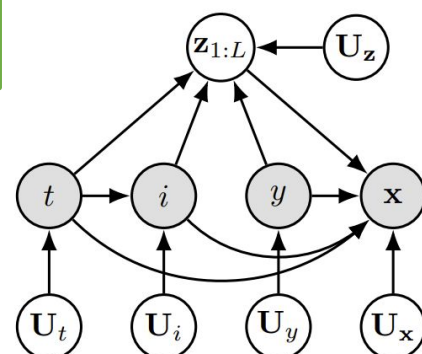
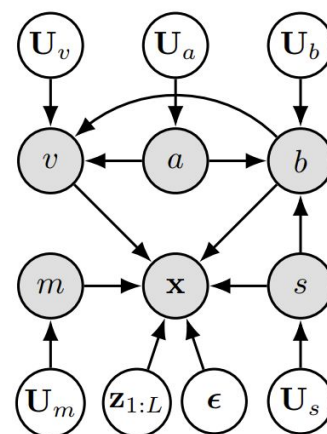
→ **High-level Example:** computing an image counterfactual



Causal Mechanism: Continuous Flow

→ **High-level Example:** computing an image counterfactual



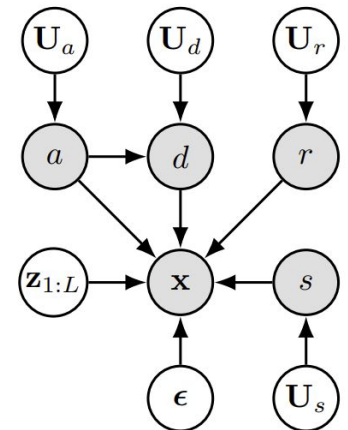
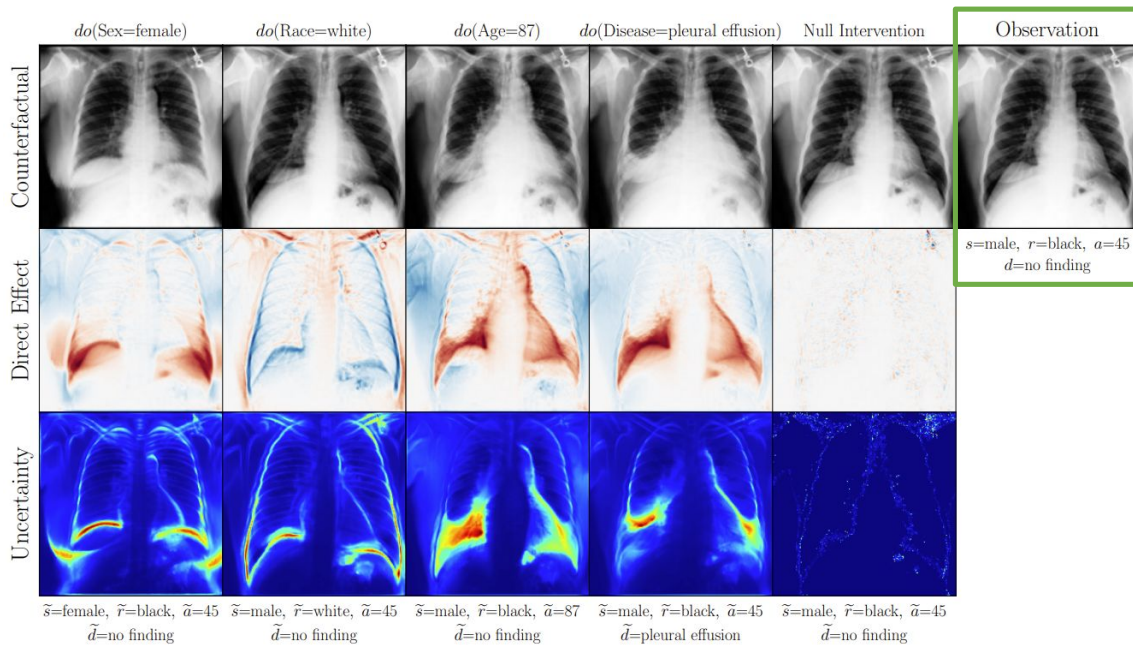
De Sousa Ribeiro, Xia, Monteiro, Pawlowski, Glocker. High Fidelity Image Counterfactuals with Probabilistic Causal Models. *ICML 2023*De Sousa Ribeiro, Xia, Monteiro, Pawlowski, Glocker. High Fidelity Image Counterfactuals with Probabilistic Causal Models. *ICML 2023*

Case Study: Chest X-ray Imaging



Hugging Face

Code & Demo



(a) Deep SCM for MIMIC-CXR. The variables in the causal graph are: age (a), sex (s), race (r), disease (d) and chest x-ray (x). The disease d is pleural effusion.

De Sousa Ribeiro, Xia, Monteiro, Pawlowski, Glocker. High Fidelity Image Counterfactuals with Probabilistic Causal Models. *ICML* 2023

Evaluating Counterfactuals: Axiomatic Properties

→ There are **3** axiomatic properties of counterfactuals:

1. Composition

2. Effectiveness

3. Reversibility

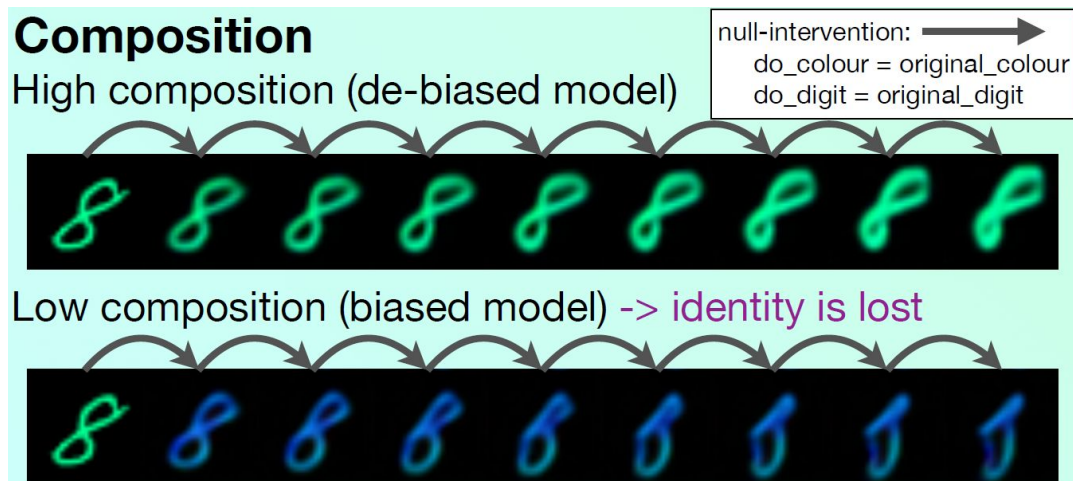
→ They are necessary and sufficient in all causal models (Galles & Pearl, 1998; Halpern, 1998)



We can evaluate counterfactuals by **measuring** these properties!

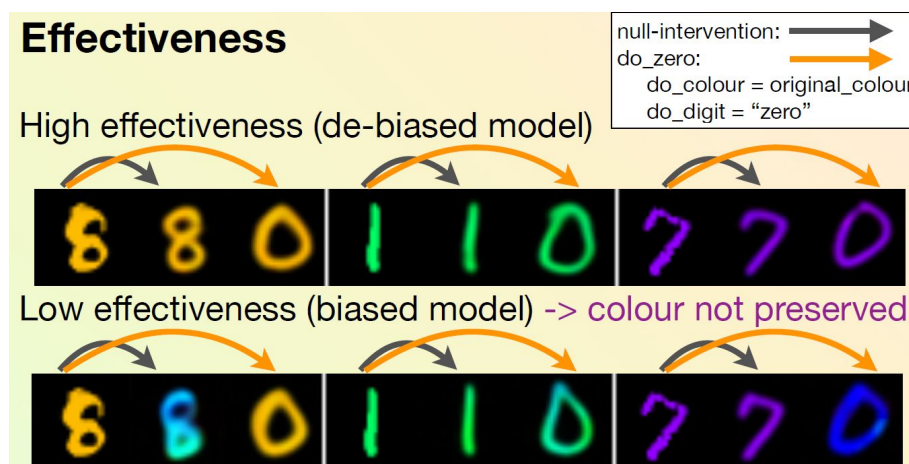
Axiom 1: Composition

Definition (informal). Intervening on a variable to have the value it would otherwise have without the intervention will not affect other variables.



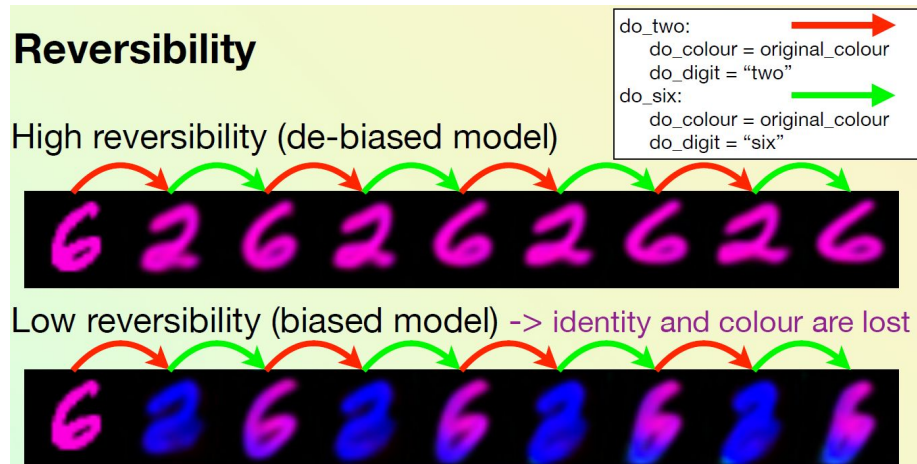
Axiom 2: Effectiveness

Definition (informal). Intervening on a variable to have a specific value will cause the variable to take on that value.



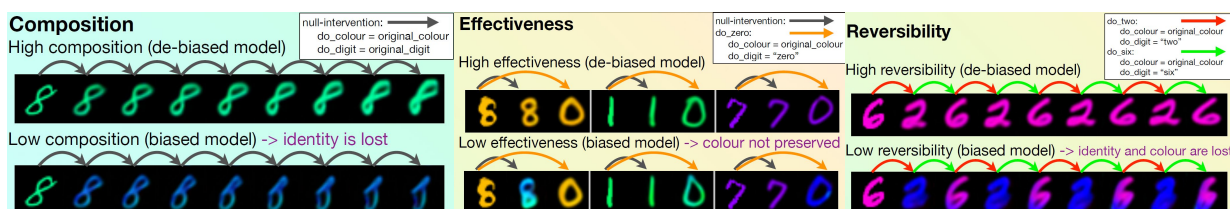
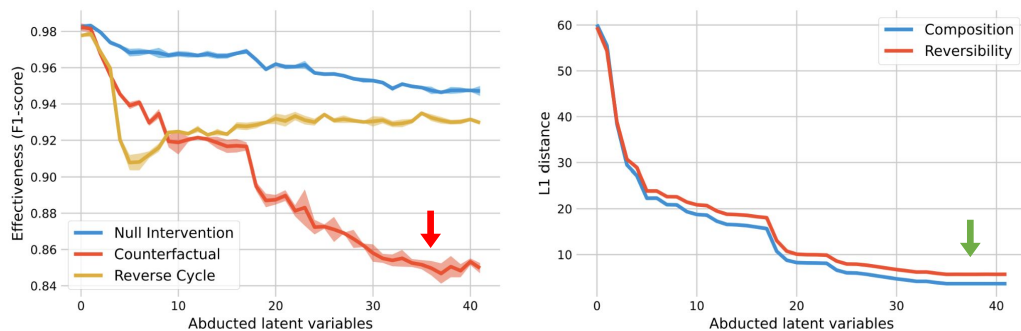
Axiom 3: Reversibility

Definition (informal). If setting a variable X to a value x results in a value y for a variable Y , and setting Y to y results in x for X , then X and Y will naturally take on the values x and y .

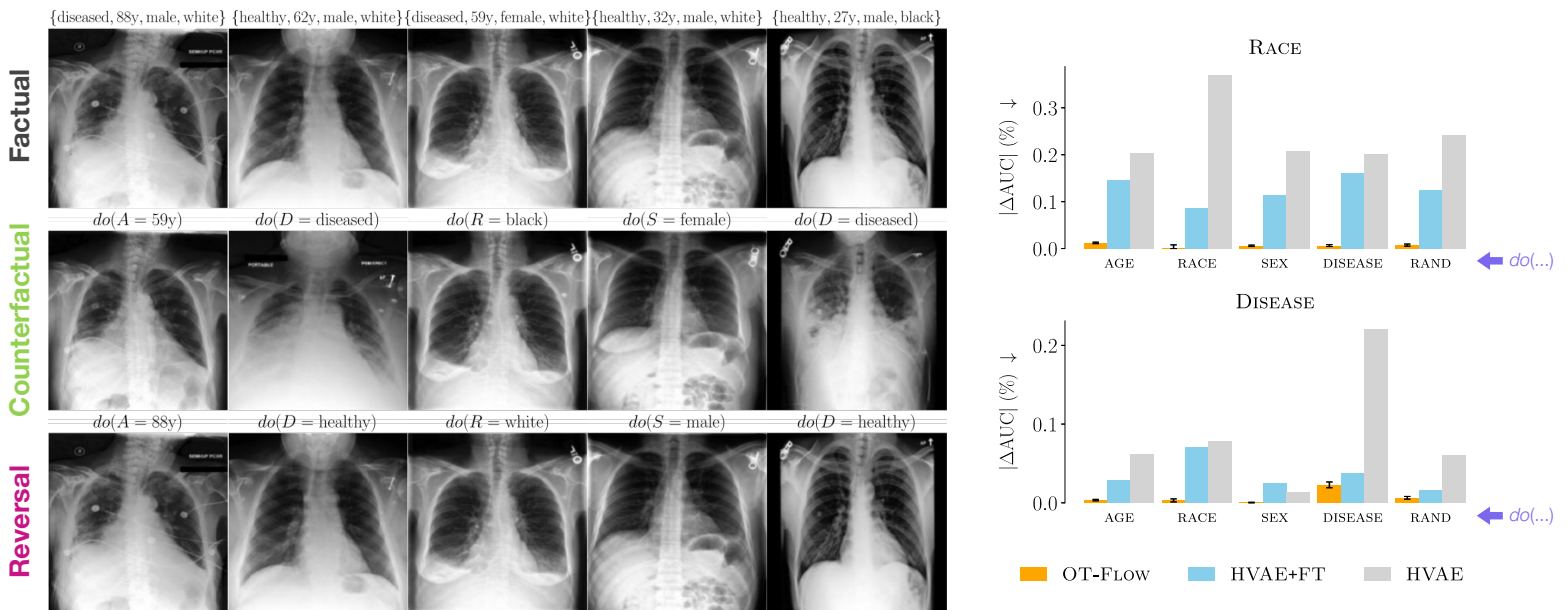


A Trade-off: Axiomatic Properties

→ **TL;DR:** We identify an inherent trade-off between the axioms



Revisit: Chest X-ray Counterfactuals



De Sousa Ribeiro, Santhirasekaram, Glocker. Counterfactual Identifiability via Dynamic Optimal Transport. *NeurIPS* 2025

Conclusion

- **Generative AI** holds great potential for **healthcare** applications
- With current technology, we can already generate plausible **counterfactuals**
- Thinking **causally** is crucial to identify and mitigate biased predictions
- Future AI systems should be aligned through **causal world models** to ensure **safe** and **reliable** decision-making

IMPERIAL



Hugging Face

Code & Demo



Thank you

Happy to take questions!



Fabio S. Ribeiro



Miguel Monteiro



Nick Pawlowski



Daniel C. Castro



Tian Xia



Ben Glocker

